

HydroSense-Kenya

Meet the team

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Kinyua

- Cleaned and processed data
- Lvl 1, 4, 6
- Slides
- Tests
- Requirements
- AI USE LOG

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- Lvl 2, 3, 5
- Presentation pdf
- src files
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- README

Project overview



Smallholder farms in Kenya often rely on intuition for irrigation, leading to either crop stress (under-watering) or wasted water and energy (over-watering). HydroSense-Kenya is a data-driven water management system that uses daily weather records and soil-sensor data from three farm zones (tomato, kale, maize) to recommend precise, minimal irrigation schedules.

Briefly describe the project objectives

01

Build a reproducible, end-to-end pipeline that turns raw sensor/weather data into optimized irrigation schedules

02

Implement and validate core numerical methods (root finding, integration, ODEs, linear systems) from scratch

03

Minimize water use while preventing crop stress, using simulation and a greedy optimization algorithm

Data Cleaning

Dataset	Column	Issue	Action	Rationale
Weather	rainfall_mm	Nan	Replaced with median	Rainfall is right-skewed; median is more robust than mean
Weather	humidity_pct	NaN	Replaced with mean	Humidity is roughly symmetric; mean is appropriate
Weather	temperature_cc	Outlier 45.8°C	Replaced with median	Median computed after excluding the faulty value to avoid self-bias
Weather	rainfall_mm	Outlier 85.0 mm	Flagged	Extreme rainfall is meteorologically possible; removing it would lose a real signal

Data Cleaning

Dataset Column		Issue	Action	Rationale
Soil	soil_moisture_pct	Nan	Replaced w/ ZONE_B median 30.7%	Per-zone median respects zone-specific soil characteristics
Soil	soil_moisture_pct	Outlier, 8.5% below threshold 24%	Replaced w/ ZONE_B median 30.7%	Consistent with a sensor misread; no corroborating weather event explains a drop this large
Soil	tank_level_litres	9900L - likely entry error	Replaced w/ ZONE_Cmedian 4054L	Value is nearly double the dataset maximum; plausible transcription error
Soil	pump_flow_lpm	Sensor_status = CHECK	Replaced w/ ZONE_B median 21.6 L/min	Zero flow with normal power draw is physically inconsistent; <code>sensor_status</code> updated to <code>IMPUTED</code>

Methods Implemented

Root Finding

Linear Systems

Integration Optimizer

- Bisection, Newton-Raphson, Secant method.
- Three methods to find where a function equals zero; used to solve the water-balance equation for equilibrium moisture.

- Gaussian elimination.
- Solves systems of linear equations; used to allocate water across multiple crop zones simultaneously.

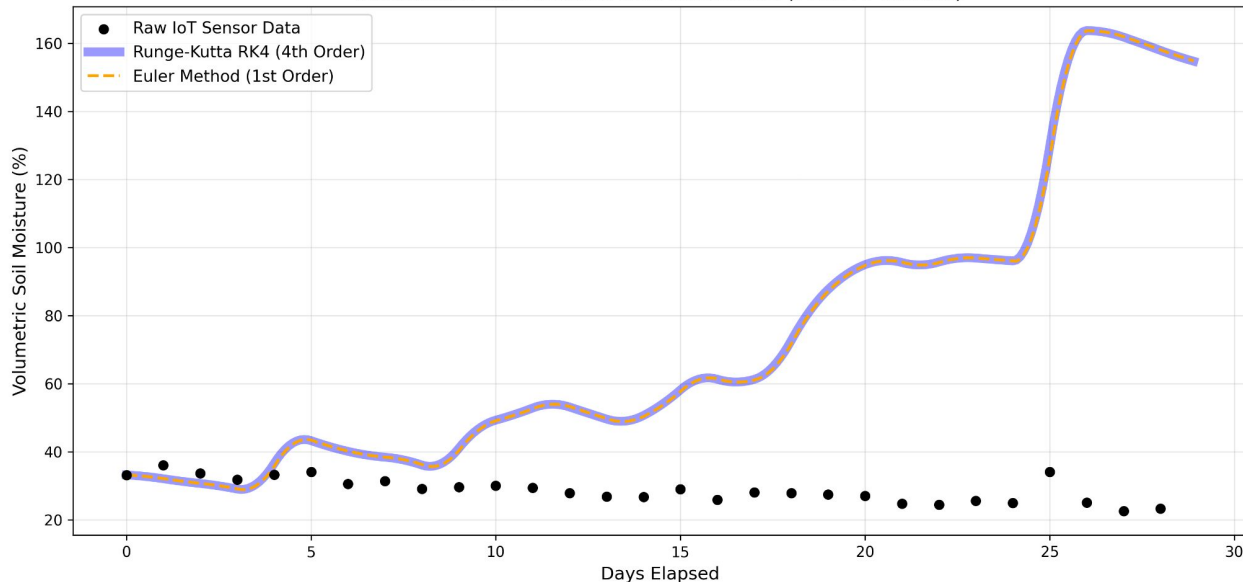
- Euler and RK4.
- Step-by-step ODE solvers that simulate how soil moisture changes each day; RK4 is more accurate than Euler.

- Greedy one-day-ahead optimizer.
- Each day, checks if tomorrow's projected moisture falls below the crop's critical threshold; if so, applies the exact minimum water needed to stay safe.

Simulation Results: Euler v RK4 plot



Continuous Soil Moisture ODE Simulation (Zone A: Tomato)



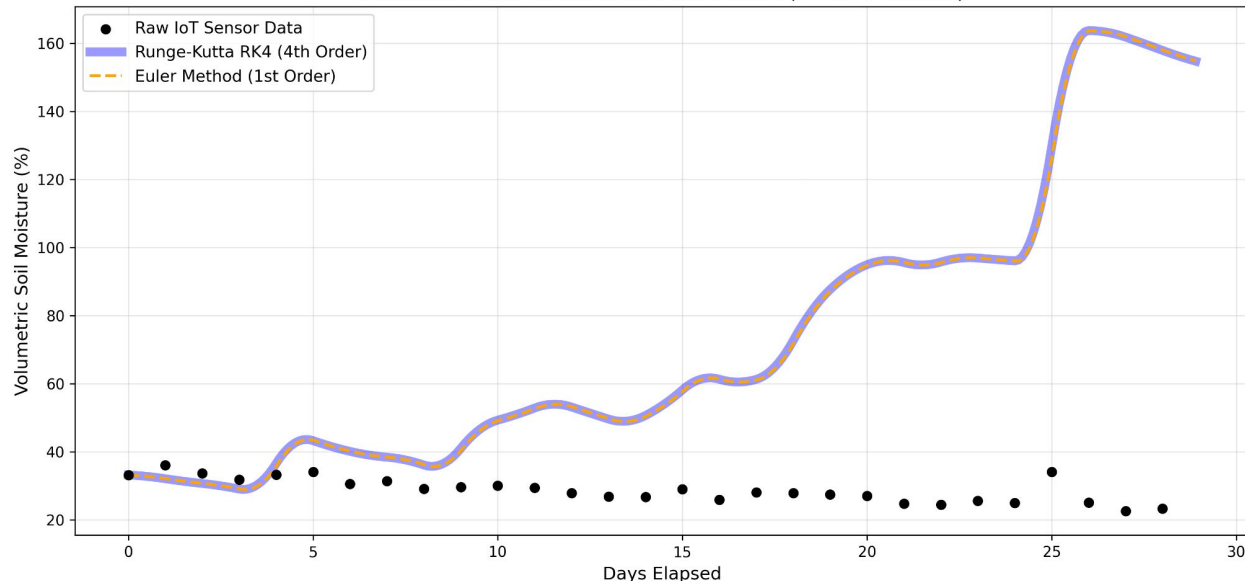
- Both Euler and RK4 track closely; RK4 is smoother and more accurate
- Moisture starts at 33.2%, rises as rainfall exceeds ET
- Spikes correspond to high rainfall days

Optimization Results:

Irrigation schedule plot



Continuous Soil Moisture ODE Simulation (Zone A: Tomato)



- Total irrigation = 0.0 mm; rainfall alone was sufficient for the 30-day window
- Model correctly identified no intervention was needed — no over-irrigation
- Any dips below the threshold would have triggered the minimum water dose automatically

Conclusion



Built a fully reproducible pipeline from raw sensor data to irrigation recommendations



All numerical methods validated against analytical solutions and cross-checked against each other



51/51 tests passing; AI use documented and reviewed at every step

Thank you!

Questions?